Mathematics

PART – III

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

35. If 7 balls are drawn one by one with replacement from a bag containing 5 balls numbered 0, 2, 3, 4, and 5, then number of ways in which a sum of 30 can be obtained, is not a multiple of
(A) 3
(B) 5
(C) 7
(D) 8

36. If $\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\left[\frac{mn}{3^n}\left(\frac{1}{3^m}-\frac{n}{n\cdot 3^m+m\cdot 3^n}\right)\right]=\frac{p}{q}$ (where p and q are relatively prime), then q – 3p is
(A) 2
(B) 3
(C) 4
(D) 5

37. Consider three sets
$A \equiv \left\{x : 2\cos 2x + 2\left(\sqrt{3}+1\right)\cos x + 2 + \sqrt{3} = 0\right\}$
$B \equiv \left\{x : 2\left(\sin 3x + \sin x\right) + \sqrt{3} = 0\right\}$
$C \equiv \left\{x : \sqrt{2}-1 < \tan x < \sqrt{2}+1\right\}$
If the sum of elements in $A \cap B \cap C$ is equal to $5k\pi$; $x \in \left[0, \frac{15\pi}{2}\right]$, then the value of k is
(A) $\frac{52}{15}$
(B) $\frac{52}{25}$
(C) $\frac{52}{5}$
(D) 3

38. Let y(x) be a solution of the differential equation $y^2\frac{dx}{dy}+x^2e^{\frac{y(x-y)}{x}}=2x(y-x)$. If y(1) = 0 and $y(x_0) = 1$, then the value of $\frac{1}{x_0}-\log_e(2e-1)$ is equal to
(A) 0
(B) –1
(C) 1
(D) $\frac{1}{2}$

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

39. If f(x), g(x) and h(x) are polynomials of degree 2 and $\Delta(x)=\begin{vmatrix} f(x) & g(x) & h(x) \\ f(x-a) & g(x-a) & h(x-a) \\ f(x-b) & g(x-b) & h(x-b)\end{vmatrix}$, then
(A) $\Delta'(x)$ is independent of x
(B) $\Delta''(x)$ is independent of x but dependent on a and b
(C) $\Delta''(x)$ is independent of x, a and b
(D) $\Delta'''(x)$ is independent of x, a and b